

Arithmetic Sequences - Answer Key

1. Yes; $d=6$
2. No; the consecutive differences are not constant.
3. Yes; $d=-5$
4. No; the consecutive differences are not constant.
5. Yes; $d=5$
6. No; the consecutive differences are not constant.
7. Yes; $d=-7$
8. No; the consecutive differences are not constant.
9. Yes; $d=0$
10. $d=-4$; next terms: 5, 1, -3
11. $d=7$; next terms: 16, 23, 30
12. $d=8$; next terms: 7, 15, 23
13. 75
14. -74
15. -78
16. 20th term
17. 18th term
18. $a_n=28-6(n-1)$
19. -7, -2, 3, 8, 13; $a_n=-7+5(n-1)$
20. $a_n=24+3(n-1)$; number of seats in row 18 is 75
21. $a_n=45+10(n-1)$; amount saved during week 12 is 155 dollars